

Question Paper

Exam Date & Time: 29-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE, MAKEUP EXAMINATIONS - DECEMBER 2025
Artificial Intelligence [CSE_3128]

Marks: 50

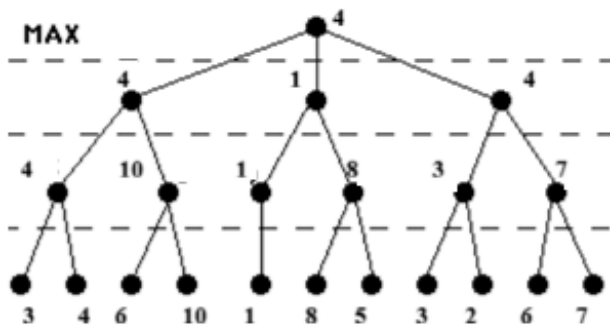
Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Analyze the efficiency and optimality of BFS and DFS for the Water Jug problem. (5)
A) Course Outcome: "CO1" / Cognitive Level: "Analysis"
B) Compare and contrast the characteristics and functionalities of different types of intelligent agents. (5)
Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 2) Analyse the different types of environments for an agent. Provide suitable examples. (5)
A) Course Outcome: "CO2" / Cognitive Level: "Analysis"
B) Provide an analysis on intelligent agent architecture considering simple reflex and memory based reflex agents and their respective environments for Cancer Diagnosis System. (5)
Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 3) How depth-limited search is different from DFS? Apply their usage along with suitable example of your choice. (5)
A) Course Outcome: "CO3" / Cognitive Level: "Analysis"
B) How applying hill climbing strategy can do efficient searching? Can you also discuss about the three limiting conditions as well? (5)
Course Outcome: "CO3" / Cognitive Level: "Application"
- 4) Provide an in-depth comparative analysis between uninformed search and informed search techniques. Also, provide at least one detailed example from each. (5)
A) Course Outcome: "CO3" / Cognitive Level: "Analysis"
B) How the agent can apply forward and backward reasoning to deal with uncertainty. Discuss in detail. (5)
Course Outcome: "CO3" / Cognitive Level: "Application"
- 5) (5)
A)

Signify the importance of alpha-beta pruning.
Analyze the following graph in this context.



Course Outcome: "CO5" / Cognitive Level: "Application"

B) Apply solution finding for the following cryptarithmic instance: (5)

SEND
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MORE

MONEY

Course Outcome: "CO1" / Cognitive Level: "Application"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

CSE_3128:Artificial Intelligence

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	15
3	3	CO 3	20
4	5	CO 5	5
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

CSE_3128:Artificial Intelligence

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	4	20(40%)
4	Analysis	6	30(60%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)